

Native precision, marked detection and quantitative kernel leakage

Split-Zero research programme

19 September 2026

This is the receiving proof for the supplied continuation [1]. It retains the original cutoff, full root polynomial, period, unit phase, minimum metrics and marked scale. Its analytic statements use the stated estimates of the preceding native proof; their finite coefficient checks are not substituted for those estimates. We prove the sharper precision consequences, the first observed marked defect, the exact stack metric, and the certified heat depth. We also derive an explicit positive lower bound for the original leakage Gram, using the actual observation inverse and both action cross blocks.

1 The precise analytic inputs and their receiving maps

On the original simple-quartet domain let

$$\begin{aligned} a &\equiv 1 \pmod{4}, & q &= (a+1)^2, & Q &= (a-7)^2, \\ \Delta &= 16a - 48, & v &= \text{ord}_0 E_A, & \epsilon &\in \{0, 1\}, & s &= 1 + \epsilon(a-1). \end{aligned} \quad (\text{NP1})$$

Set $\ell = (s-1)/4$, $n = Q + \ell$, $m = \Delta - v$ and $H = 2\Delta - v - \ell$. These retain the original source order even for the degree- $(a-8)$ lower root grid. Its four cutoffs are $q - v - 1$, $q - v$, $2q - v - 1$, $2q - v$.

Use the actual equilibrium endpoints u_e^2, v_e^2 for $\pi\sqrt{x} - 2\log x$, satisfying $u_e K(\kappa) = 2$, $v_e E(\kappa) = 4$, $\kappa^2 = 1 - u_e^2/v_e^2$. Define

$$\begin{aligned} w &= (u_e + v_e)/2, & z_e &= u_e v_e, & c_e &= (v_e^2 - u_e^2)/4, \\ L &= 6\log w - 2\log z_e - 2, & \lambda_e &= 8 - 4\log w - 2\log c_e, & F &= -\lambda_e/2 - 3 + L, \\ C_B &= 9 - 8\log 2 + F, & C_\Gamma &= 2L - 8\log 2 + 4, \\ C_\sigma &= F + 6 - 3L/2 - 6\log 2 + 2\log \pi, & M_{-1} &= (u_e^2 + v_e^2 - 4u_e v_e)/(2u_e^2 v_e^2), \\ a_0 &= \log(4/\pi), & a_1 &= 2(1 - \log 2) - \lambda_e/4, & b_1 &= \frac{1}{2}\log c_e - \log 2, \\ C_\partial &= 2C_B - 8a_1 + 4a_0, & K_2 &= C_B - 4a_1 + 2b_1 - 2\log 2, & \eta_h &= (\gamma^2 - \delta^2)/3. \end{aligned} \quad (\text{NP2})$$

The quantitative input D17 of [2] is

$$\begin{aligned} C_{a,s}^- &= C_B n^2 + 4n(Ha_1 - ma_0) + \frac{1}{2}m^2 \log(n/m) + 2H^2 b_1 \\ &\quad - (2a_0 - \frac{3}{4})m^2 - 2\ell^2 \log 2 + C_\sigma n - 2\eta_h Q(Q-1)M_{-1}/n + O_{h,v}(a^{3/2} + a \log(a+2)). \end{aligned} \quad (\text{NP3})$$

Its proof sums the low rows with their full endpoint weights, including $\sum c_r r = m^2$ and $\sum c_r r \log r = m^2 \log m - m^2/2 + O(\log(m+2))$. The expansion $\psi(t) = a_0 + (t/4)\log t + (a_0 - 1/4)t + O(t^{3/2})$ gives the $-3/4$ in (NP3): $-1/2$ from the linear summand and $-1/4$ from this weighted logarithmic sum. No interval is omitted.

Theorem 1. *Let*

$$Z(v) = 1024(a_1 - 2b_1) - 176C_\partial + 4v(a_1 - a_0) - 192 + 512\log 2, \quad Z_\Delta = 4C_\partial + 16(2b_1 - a_1). \quad (\text{NP4})$$

The expanded lower covariance satisfies

$$\begin{aligned} \mathcal{C}_{a,s}^- &= C_B q^2 + [-16C_\partial + \epsilon C_\Gamma/4]aq + 128q \log a \\ &\quad + q[C_\sigma - 2\eta_h M_{-1} - \epsilon C_\Gamma/4 + \epsilon K_2/16 - Z(v) - \epsilon Z_\Delta] \\ &\quad + O_{h,v}(a^{3/2} + a \log(a+2)). \end{aligned} \quad (\text{NP5})$$

The complete original correlated native sum obeys

$$\mathcal{N}_{a,s} = 16C_\partial aq - 128q \log a + [Z(v) + \epsilon Z_\Delta]q + O_{h,v}(a^{3/2} + a \log(a+2)). \quad (\text{NP6})$$

After the displayed terms, each remainder is $o(q/\log q)$; in particular its next $q/\log q$ coefficient is zero. Also $\mathcal{N}_{a,a} - \mathcal{N}_{a,1} = Z_\Delta q + o(q/\log q)$.

Proof. Substitute $n = a^2 + (-14 + \epsilon/4)a + 49 - \epsilon/4$ into (NP3). All polynomial terms of degrees four, three and two yield (NP5) by expansion. In the logarithmic term, $\log(n/m) = \log a - \log 16 + O_{h,v}(a^{-1})$, so its discarded part is $O(a)$; changing its degree-two coefficient from a^2 to q costs $O(a \log a)$. Further, $Q(Q-1)/n = q + O_{h,v}(a)$ and $n = q + O(a)$. These estimates account for every nonpolynomial replacement. The supplied exact coefficient diagnostics independently check the three polynomial coefficients for both $\epsilon = 0, 1$.

The original receiving identity is $\mathcal{B}_a^{(s)} = \mathcal{C}_{a,s}^- + \mathcal{N}_{a,s}$, where $\mathcal{N} = \mathcal{R}e^C + \mathcal{T} + F_L + \Xi$ retains all four specified terms. The upper Gamma baseline is

$$\mathcal{B}_a^{(s)} = C_B q^2 + \epsilon C_\Gamma aq/4 + q[C_\sigma - 2\eta_h M_{-1} - \epsilon C_\Gamma/4 + \epsilon K_2/16] + O_h(a \log(a+2)).$$

Its order-one remainder is $O_h(\log q)$ in the included absolute baseline proof; D20's order conversion costs $O_h(a \log q)$. Subtracting proves (NP6). Finally $(a^{3/2} + a \log(a+2)) \log q/q \rightarrow 0$. The claims concern this complete correlated sum and covariance, not their individual restriction or quotient determinants. $\square$

At fixed original multiplicity m_0 , retain $q = [1 + k(m_0 - 1)](k+1)^2$, $\ell = (k-1)/4$ and the original arithmetic density. D20 gives the exact source-order comparison through its stated error. With the earlier arithmetic result SR/AG as used in FW29–30 [3], subtraction yields

$$\begin{aligned} \delta_{k,k}^{\text{ar}} &= -(\ell + 4m_0 - 2)C_\Gamma q + \frac{\log 2}{\pi} I_h k^2 - \ell^2 K_2 + \frac{C_\Gamma q}{2(\log q + c_\zeta)} + o_h(q/\log q), \\ I_h &= \int_0^{\pi/2} [(\log M_h)'(t)]^2 dt, \quad M_h(t) = \int_{\mathbb{R}} e^{ty} w_h(y) dy, \quad c_\zeta = 2\gamma_E - \log(2\pi). \end{aligned} \quad (\text{NP7})$$

Here $w_h(y) = |2\xi(1/2 + iy)/h(1/2 + iy)|^2/(2\pi)$ and h is the complete fixed quartet polynomial. The factor k^2 remains literal. For $m_0 > 1$ an $o(q/\log q)$ remainder does not determine every smaller k^2 -scale term separately. The proof is subtraction of $\ell q C_\Gamma + \ell^2 K_2 + O_h(k \log q)$ from the order-one correction; $k(\log q)^2/q \rightarrow 0$ proves the error claim. The exact complete-action receiver is

$$\mathcal{B}_k[w_k] + \delta_{k,k}^{\text{ar}} = \mathcal{B}_k[\sigma] + \delta_{k,1}^{\text{ar}}. \quad (\text{NP8})$$

The larger conversion terms cancel in this identity.

For the sign in D21, set $r = u_e/v_e$ and $\kappa = \sqrt{1 - r^2}$. Differentiating the elliptic integrals using Carlson's formulas [6, 19.4.1–2] gives exactly

$$\frac{d}{dr} \frac{rK(\kappa)}{E(\kappa)} = \frac{(K - E)(E - r^2 K)}{(1 - r^2)E^2} > 0. \quad (\text{NP9})$$

The first factor is positive by the integral of $\kappa^2 \sin^2 t / \sqrt{1 - \kappa^2 \sin^2 t}$; the second by the integral of $\kappa^2 \cos^2 t / \sqrt{1 - \kappa^2 \sin^2 t}$. For completeness, the substitution $x = \tan t$ bounds $K = \int_0^\infty [(1 + x^2)(1 + r^2 x^2)]^{-1/2} dx$ by $1 + \log(1/r) + 1$, using the intervals $[0, 1]$, $[1, 1/r]$, and $[1/r, \infty)$. Also $E \geq \int_0^{\pi/2} \cos t dt = 1$. At $r = 1/10$, $rK/E < (2 + 3)/10 = 1/2$, so the actual solution of $rK/E = 1/2$ exceeds $1/10$. Since $1 < (1 + r)^2/(4r) < 121/40$, the exact identity $K_2 = 2 \log(w^2/z_e) - 4 \log 2$ yields $-4 \log 2 < K_2 < 2 \log(121/160) < 0$.

2 The first visible jet and the original marked recurrence

For this section use the original simple quartet and $k \geq 9$, $k \equiv 1 \pmod{4}$. Retain its actual invertible period matrix $H(\varpi)$, and put

$$x(z) = H(\varpi)(e^{2(\delta+i\gamma)z}, e^{2\delta z}, e^{2i\gamma z}, 1)^T, \quad b_* = 1/2 - \delta - i\gamma. \quad (\text{NP10})$$

The original observation pairs $P(S)$ with $P(\partial_z)[e^{kb_*z}f(x(z))]|_{z=0}$, for invariant homogeneous f of degree k with coordinate weights $1, 2, 3, 4$ modulo five. Define $j_* = \min\{j : \Lambda S^j \neq 0\}$.

Theorem 2. *If $x(0)$ has two active coordinates then $j_* = 0$. If only $x_i(0)$ is active, write $z_j = x_j/x_i$, $\nu_j = \text{ord}_0 z_j$ and $c_j = j - i \pmod{5}$. Then*

$$j_* = \min_{\substack{\alpha \geq 0, |\alpha| \leq 4 \\ \sum_{j \neq i} c_j \alpha_j = -ki \pmod{5}}} \sum_{j \neq i} \nu_j \alpha_j \leq 4. \quad (\text{NP11})$$

In this case $j_ = 0$ exactly when $5 \mid k$.*

Proof. If coordinates i, j are active, choose $0 \leq b \leq 4$ with $ki + b(j - i) = 0 \pmod{5}$. The monomial $x_i^{k-b}x_j^b$ is invariant and nonzero at zero. In the one-active case, each inactive coordinate is a nonzero four-exponential sum: its row in H is nonzero. The first four exponential derivatives give an invertible Vandermonde, so $1 \leq \nu_j \leq 3$. At least one equals one, since otherwise $x'(0)$ is proportional to $x(0)$, which invertibility of H would make the distinct exponent vector proportional to $(1, 1, 1, 1)$.

Divide each invariant monomial by the common unit $e^{kb_*z}x_i(z)^k$. Its charge is exactly that in (NP11), and its order is the indicated sum. A linear combination cannot have smaller order than every summand. Conversely each candidate of degree at most four homogenizes to degree k and has that exact order. A power at most four of an order-one ratio realizes every charge; any monomial of total degree at least five has order at least five and cannot improve the minimum. The charge-zero constant realizes order zero; all other charges have positive order. This proves every assertion. $\square$

Let $E = \mathbb{C}[S]/(\chi)$, $\deg \chi = q$, in the original increasing power basis. Denote multiplication by S by A , and let $B = e_0 e_{q-1}^T$. Retain the original marked scale $u \neq 0$ and the connection

$$A(t) = A + tB, \quad U'(t) = A(t)U(t)/u, \quad U(0) = I, \quad \Lambda_h(t) = \Lambda U(t)^{-1}. \quad (\text{NP12})$$

For the actual terminal class $v_\lambda = b_\lambda \chi / (S - \lambda)$, one has $Av_\lambda = \lambda v_\lambda$ and $Bv_\lambda = b_\lambda 1$. Thus, with $Y(t) = e^{\lambda t/u} \Lambda_h(t) v_\lambda - \Lambda v_\lambda$,

$$Y'(t) = -\frac{b_\lambda}{u} t e^{\lambda t/u} \Lambda_h(t) 1. \quad (\text{NP13})$$

This follows by differentiating U^{-1} on its correct side: $(U^{-1})' = -U^{-1}A(t)/u$.

To calculate the first coefficient without an unjustified commutation, write $U^{-1} = \sum W_n t^n$. Then $(n+1)W_{n+1} = -(W_n A + W_{n-1} B)/u$, $W_{-1} = 0$. In any word contributing to W_n , the letter A has degree one and B degree two. A word containing B kills 1 unless its rightmost B has at least $q-1$ letters A to its right, since $BA^j 1 = 0$ for $j < q-1$. Hence $W_n 1 = (-A/u)^n 1/n!$ for $n < q+1$. Since $j_* \leq 4 < q$, (NP13) proves

$$Y^{(j_*+2)}(0) = b_\lambda(j_*+1)(-u)^{-(j_*+1)} \Lambda S^{j_*} \neq 0, \quad Y^{(j)}(0) = 0 \quad (j < j_*+2). \quad (\text{NP14})$$

The first marked/arithmetic discrepancy is therefore at order two on the two-active stratum, and at an exactly determined order at most six at every admitted period. This computes the stated holomorphic continuation, with full b_λ and u ; it does not substitute a new sheaf morphism for that continuation.

3 Observation stacks in their actual quotient metric

Fix an original cutoff N , its positive form $G = G_N$, and the surjective observation $\Lambda : E \rightarrow B_0 = \text{im } \Lambda$. Define

$$Q = (\Lambda G^{-1} \Lambda^*)^{-1}, \quad L = G^{-1} \Lambda^* Q, \quad K = \ker \Lambda, \quad I : \mathbb{C}^r \longrightarrow K, \quad H_K = I^* G I. \quad (\text{NP15})$$

I is the original fixed full-column kernel frame. Thus $\Lambda L = I$ on B_0 , $I^* G L = 0$ and $L^* G L = Q$. Set $J_K = H_K^{-1} I^* G$, so $I J_K + L \Lambda = I_E$. The literal action blocks are

$$T = J_K A I, \quad C = J_K A L, \quad B = \Lambda A I, \quad D = \Lambda A L, \quad A[I, L] = [I, L] \begin{pmatrix} T & C \\ B & D \end{pmatrix}. \quad (\text{NP16})$$

All norms below use H_K on kernel coordinates and Q on outputs.

The incoming O1–9 theorem [2] proves $K_\infty = 0$ on its explicit original period locus, with $r = 8k - 16$, and injectivity of $\mathcal{O}_r = (\Lambda, \Lambda A, \dots, \Lambda A^r)^T$. Its proof is the decreasing-kernel argument: a plateau is invariant under A , hence zero; at most r strict drops start from K . Consequently

$$\mathcal{L}_r = \sum_{j=0}^{r-1} (T^j)^* B^* Q B T^j \succ 0. \quad (\text{NP17})$$

Indeed if all $B T^j x = 0$ for $j < r$, the characteristic polynomial of T gives the same for every j . The block recurrence then gives $A^j I x = I T^j x$, so $I x \in K_\infty$ and $x = 0$.

The full stack metric is

$$H_{d,N} = \sum_{j=0}^d (A^j)^* \Lambda^* Q \Lambda A^j, \quad v_\lambda^* H_{d,N} v_\lambda = \left(\sum_{j=0}^d |\lambda|^{2j} \right) E_{N,\lambda} (1 - \theta_N), \quad (\text{NP18})$$

where $E_{N,\lambda} = v_\lambda^* G_N v_\lambda$ and $\theta_N = \|J_K v_\lambda\|_{H_K}^2 / E_{N,\lambda}$. The equality is substitution of the actual eigenvalue equation into each term. For fixed d the geometric factor cancels in the original four-cutoff return; FW38–40 [3] therefore yields

$$C_{\partial q} + \log \frac{(1 - \theta_{q-1})(1 - \theta_q)}{(1 - \theta_{2q-1})(1 - \theta_{2q})} + O_h(k + \log q). \quad (\text{NP19})$$

The stack retains the original angle term by an exact identity.

4 A quantitative lower bound for every kernel direction

We now extend positivity (NP17) by a bound expressed in actual finite matrices. This is valid on the just-stated simple-packet observability locus and at every original cutoff. It does not replace that cutoff.

Let the distinct arithmetic roots be $\lambda_1, \dots, \lambda_q$, with original Lagrange idempotents $e_\lambda(S) = \prod_{\mu \neq \lambda} (S - \mu) / (\lambda - \mu) = \sum_{j=0}^{q-1} e_{\lambda,j} S^j$. Write $b_\lambda^{\text{obs}} = \Lambda e_\lambda \neq 0$ and define

$$\vartheta_\lambda(y) = \frac{(b_\lambda^{\text{obs}})^* Q y}{(b_\lambda^{\text{obs}})^* Q b_\lambda^{\text{obs}}}, \quad \mathcal{R}_N(y_0, \dots, y_{q-1}) = \sum_\lambda e_\lambda \sum_{j=0}^{q-1} e_{\lambda,j} \vartheta_\lambda(y_j). \quad (\text{NP20})$$

For the original stack, its λ coordinate is $\vartheta_\lambda(\Lambda e_\lambda(A)x) = x_\lambda$, so $\mathcal{R}_N \mathcal{O}_{q-1} = I_E$. This is O10 with its functional explicitly placed in the original quotient metric Q ; no metric equivalence is silently assumed. By Cauchy–Schwarz,

$$\|\mathcal{R}_N\|_{Q^{\oplus q} \rightarrow G} \leq C_{R,N} := \sum_\lambda \frac{\|e_\lambda\|_G (\sum_j |e_{\lambda,j}|^2)^{1/2}}{\|b_\lambda^{\text{obs}}\|_Q} < \infty. \quad (\text{NP21})$$

This coefficient contains the original root differences and metric.

For $x \in \mathbb{C}^r$ write $y_j = \Lambda A^j I x$, $x_j = J_K A^j I x$. The exact block recurrences are $x_{j+1} = T x_j + C y_j$, $y_{j+1} = B x_j + D y_j$, with $x_0 = x, y_0 = 0$. Elimination of x_j gives

$$y_{j+1} = B T^j x + D y_j + \sum_{l=0}^{j-1} B T^{j-1-l} C y_l. \quad (\text{NP22})$$

Define a $(q-1)$ by $(q-1)$ block lower-triangular matrix W_N by the following recursion. For input $z_0, \dots, z_{q-2}$ set $y_0 = 0$ and $y_{j+1} = z_j + D y_j + \sum_{l=0}^{j-1} B T^{j-1-l} C y_l$. Then $W_N z = (y_1, \dots, y_{q-1})$. Its diagonal blocks are identities; the displayed reverse formula gives its exact inverse. In particular, both action cross blocks B, C and D are retained.

Let $\chi_T(t) = \det(tI - T)$, and compute the actual polynomial remainders $t^j \bmod \chi_T = \sum_{h=0}^{r-1} c_{jh} t^h$, $0 \leq j \leq q-2$. Put $C_T = (c_{jh})$. The first r rows form the identity. Applying the characteristic identity to T proves

$$(B T^j x)_{j=0}^{q-2} = (C_T \otimes I_{B_0})(B T^h x)_{h=0}^{r-1}, \quad \mathcal{O}_{q-1} I x = (0, W_N(C_T \otimes I_{B_0})(B T^h x)_{h=0}^{r-1}). \quad (\text{NP23})$$

Define $C_{W,N} = \|W_N\|_{Q^{\oplus(q-1)}}$ and $C_{T,N} = \|C_T\|_2$. These are norms of the displayed finite matrices, not constants of a different stack. Then

$$\boxed{\mathcal{L}_r \succeq (C_{R,N} C_{W,N} C_{T,N})^{-2} H_K.} \quad (\text{NP24})$$

To prove it, apply (NP20) to (NP23), take the original G norm, and then use (NP21) and the two matrix norms. The square of $\|(B T^h x)_{h < r}\|_{Q^{\oplus r}}$ is exactly $x^* \mathcal{L}_r x$; $\|I x\|_G^2 = x^* H_K x$. Rearrangement gives (NP24). It supplies a strictly positive finite lower bound at the original cutoff. Its dependence on the actual Q and action blocks is displayed. No small asymptotic action allowance is inferred from this possibly large recovery cost.

5 The exact heat guard and its evaluated degree cost

Retain HR1–11 [4], including the original complete prequotient constant $c_N \geq 1$ and the same vectors and observation. Let $0 < p_{1,N}, p_{2,N} \leq 1$ be the actual squared source-correlation fractions from FW5, and $p_{*,N} = \min(p_{1,N}, p_{2,N})$. For $0 < \zeta < 1$, the certificate

$$\eta_{J,N} = (1 + 2^{-(J+1)/2} c_N)^{2k} - 1 \leq \zeta \sqrt{p_{*,N}} \quad (\text{NP25})$$

is equivalent, by monotonicity, to

$$J + 1 \geq T_N := 2 \log_2 c_N - 2 \log_2 \left[(1 + \zeta \sqrt{p_{*,N}})^{1/(2k)} - 1 \right]. \quad (\text{NP26})$$

The least nonnegative integer depth is $J_N^* = \lceil T_N \rceil - 1$. Indeed $c_N \geq 1$ and the bracket is less than one; for the original $k \geq 9$, $T_N > 1$. There is therefore no hidden cutoff at depth zero.

For $0 < x \leq \zeta$ put $b = \log(1+x)/(2k)$. The elementary bounds $x/(1+\zeta) \leq \log(1+x) \leq x$ and $b \leq e^b - 1 \leq (1+\zeta)^{1/2} b$ show, uniformly even as $p_{*,N} \downarrow 0$, that

$$J_N^* + 1 - 2 \log_2 c_N = 2 \log_2(2k/\zeta) - \log_2 p_{*,N} + O_\zeta(1). \quad (\text{NP27})$$

Here the equality concerns the least certified depth, not an arbitrary larger J . The rounding error is less than one.

Let the original endpoint weights be $w_0 = w_q = 1$, $w_r = 2$ otherwise. They sum to $2q$. FW27–29 supplies $\mathcal{A}_1 = -\sum w_r \log p_{1,q-1+r} = C_B q^2 + O_h(q \log q)$ and $\mathcal{A}_2 = O_h(q \log q)$. Since

$$-\log p_{1,N} \leq -\log p_{*,N} \leq -\log p_{1,N} - \log p_{2,N},$$

weighted summation of (NP27) proves

$$\frac{1}{2q} \sum_{r=0}^q w_r [J_{q-1+r}^* + 1 - 2 \log_2 c_{q-1+r}] = \frac{C_B}{2 \log 2} q + O_{h,\zeta}(\log q). \quad (\text{NP28})$$

On the simple-quartet domain FW48 at $r = 0$ gives $-\log p_{1,q-1} = 2a_0 q + O_h(k + \log q)$ and $-\log p_{2,q-1} = O_h(\log q)$. The same two-sided inequality yields

$$J_{q-1}^* + 1 - 2 \log_2 c_{q-1} = \frac{2 \log(4/\pi)}{\log 2} q + O_{h,\zeta}(k + \log q). \quad (\text{NP29})$$

(NP28) uses the all-fixed-multiplicity source products; (NP29) uses the stated simple-packet pointwise estimate. These are costs of this particular heat certificate. They do not bound every possible numerical method or evaluate the separate growth of c_N .

6 Return to the now-proved coprime proper-source domain

The latest published NC1–27 proof [5] establishes the coprime stratum required in Section 5 of the incoming continuation: outside a specified finite exceptional set of admitted periods, $E_k^{\text{prop}} = 0$, with a sufficient original finite-period radius. On its intersection with the two-active jet locus, $r_I = 8k - 32 - v$. At the same proper-source cutoffs the source form is exactly $Q_N^p = Z_R^* J_0^* D_{k,N-v}^{(s_0)} J_0 Z_R$.

In the full target product let F_R be the actual multiplier lift, J_* the entire target-invariant direction matrix and $\mathcal{D}_N^p$ the unchanged target form. Define

$$A_N = F_R^* \mathcal{D}_N^p F_R, \quad C_N = J_*^* \mathcal{D}_N^p J_*, \quad B_N = J_*^* \mathcal{D}_N^p F_R, \quad T_N = C_N^{-1/2} B_N A_N^{-1/2}. \quad (\text{NP30})$$

The symbol T_N here denotes this displayed target cross map, not the real heat threshold in (NP26). Completing the square in the entire target fiber gives $H_N^p = A_N - B_N^* C_N^{-1} B_N = A_N^{1/2} (I - T_N^* T_N) A_N^{1/2}$. Thus the original proper-source return is exactly

$$\mathcal{M}^p = \mathcal{R}_{\text{end}} \log \det[(Q_N^p)^{-1/2} A_N (Q_N^p)^{-1/2}] + \mathcal{R}_{\text{end}} \log \det(I - T_N^* T_N). \quad (\text{NP31})$$

This determinant identity follows by multiplying determinants, not by asserting the matrices commute. The actual positive target minimum gives $I - T_N^* T_N \succ 0$. None of the covariance, observation-stack or heat-depth results evaluates its eigenvalues. They now supply sharper original inputs and a finite positive leakage bound for the continuing calculation of this same target metric.

References

- [1] Supplied web continuation, *The new continuation removes more uncertainty than its concluding summary states*, 19 September 2026, equations (1)–(23). The complete original Markdown is retained with this edition.
- [2] *Programme Native Continuation*, four complete notes supplied 19 September 2026, D1–25, revised J1–6 including J3a–b, E1–4, O1–12. Published with the NC edition below.
- [3] *Full-window source products and the retained class*, FW1–52, 19 September 2026, public commit `3c44de632b2c5ead13b23619d646803d2bbffc19`; full proof `FULL_WINDOW_SOURCE_PRODUCTS.tex` included.
- [4] *Heat approximation errors for the original complex observation responses*, HR1–11, and the full original heat metric proof HM1–30, published in commit `607728ee3c35be98720d548794727670b1946c91`.

- [5] *Vanishing of the original proper-source kernel through conductor coprimality*, NC1–27, 19 September 2026, commit `12dd3645229bfebbf1cb8f5968ecffd4dfff1e9f`.
- [6] Bille C. Carlson, *Elliptic Integrals*, Chapter 19 of the NIST Digital Library of Mathematical Functions, Section 19.4, equations 19.4.1–2. The derivative in (NP9) is derived from these formulas; its sign and the subsequent inequalities are proved from the defining integrals.